

Random Cluster Model on Regular Graphs

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Abstract

Random cluster model is a generalization of bond percolation with dependent edges/bonds. The study of the random cluster model's partition function, on graph sequences provides crucial insights into phase transitions in the same. This project report explores the asymptotic behavior of the pressure function for sequences of random d -regular graphs, showing that the limit exists and can be identified as the solution to a maximization problem. A central part of this is the approximation of the random cluster model by a 2-spin model and proving the equivalence of the limiting behavior of the corresponding pressure functions.

1. INTRODUCTION

In statistical physics models, such as Ising, Potts, random cluster, it is often insightful to look at the “Pressure function”. In probability theory, we study these models on a wide class of graphs. Here we consider the random cluster model, which is a generalization of bond percolation with dependent edges/bonds (see [3], [4] for details). Often on a graph sequence, we study the properties of the pressure function (such as differentiability, continuity etc.) to infer about the model in general. Here, we study a sequence of random regular graphs and the partition function and pressure function of the random cluster model on them. This has applications not only in physics but also in synthetic and real-world network data analysis.

The report is organized as follows. In Section 2, we introduce the notations and definitions used and state our main result. Section 3 is dedicated to the so-called rank 1 and rank 2 approximation to the random cluster model. In Section 4, we take digressions to subgraph counting polynomials and belief propagation, respectively. Then we study the convergence of the pressure function of the random cluster model on a sequence of d -regular graphs and conclude the proof of the main result in the final section.

2. BASIC DEFINITIONS AND THE MAIN RESULT

In this section, we introduce the definitions and notations that we are going to use.

Definition 1 (Random Cluster Model). *For a finite graph $G = (V, E)$, we assign to a specific edge-subset A , of open edges, the probability proportional to $q^{k(A)}w^{|A|}$, where, $k(A)$ is the number of connected components of the graph (V, A) . The partition function of the random cluster model is defined to be the normalizing constant, i.e.,*

$$Z_G(q, w) = \sum_{A \subseteq E} q^{k(A)} w^{|A|}.$$

Definition 2 (r -Spin Model). *We call a model a spin model with r -spins if there is an $r \times r$ symmetric matrix N , and a vector $\underline{\mu} \in \mathbb{R}^r$ such that for a finite graph $G = (V, E)$ the probability $\sigma : V \rightarrow \{1, 2, \dots, r\}$ is*

$$P(\sigma) = \frac{1}{Z_G(N, \underline{\mu})} \prod_{v \in V} \underline{\mu}_{\sigma(v)} \prod_{(u, v) \in E} N_{\sigma(u), \sigma(v)},$$

and the partition function is defined as

$$Z_G(N, \underline{\mu}) = \sum_{\sigma: V \rightarrow [r]} \prod_{v \in V} \underline{\mu}_{\sigma(v)} \prod_{(u, v) \in E} N_{\sigma(u), \sigma(v)}.$$

Remark. If q is an integer, when $r = q$, we call the model to be a q -spin model. It can be shown that there is a coupling (see page 9, [4]) between the q -spin model with $N = wI_q + J_q$, $\underline{\mu} = \mathbb{1}_q$, where J_q is the $q \times q$ matrix with all entries 1, and $\mathbb{1}_q$ is the vector with all entries 1, and the random cluster model with parameters q and w . In general, when the $\underline{\mu}$ will

be the vector with all entries 1, then we shall denote the model partition function $Z_G(N, \underline{\mu})$ simply with $Z_G(N)$.

The main result of the report is as follows:

Theorem 1 ([1]). *If $(G_n)_n$ is a sequence of random d -regular graphs on n vertices, then the following limit exists (in a.s. sense) and is identified as follows:*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln Z_{G_n}(q, w) = \ln \Phi_{d,q,w}$$

for any $q \geq 2$ and $w > 0$.

$$\begin{aligned} \Phi_{d,q,w} = \max_{t \in [-\pi, \pi]} & \left[\left(\sqrt{1 + \frac{w}{q}} \cos(t) + \sqrt{\frac{(q-1)w}{q}} \sin(t) \right)^d \right. \\ & \left. + (q-1) \left(\sqrt{1 + \frac{w}{q}} \cos(t) - \sqrt{\frac{w}{q(q-1)}} \sin(t) \right)^d \right] \end{aligned}$$

We shall see how this trigonometric objective function comes in later sections. The above result is also true for a sequence of “essentially large girth” graphs, which are defined below.

Definition 3 (Essentially large girth). *Call a sequence of graphs (G_n) with the number of vertices $v(G_n) \rightarrow \infty$, essentially large girth sequence of graphs if*

$$\text{for all positive integer } g, \lim_{n \rightarrow \infty} \frac{L(G_n, g)}{v(G_n)} = 0$$

where $L(G, g)$ denotes the number of cycles in G of the size at most $(g-1)$.

The remaining part of the report is dedicated to the proof of Theorem 1.

3. APPROXIMATIONS FOR THE PARTITION FUNCTION

To calculate some limit involving the pressure function, it can be a useful strategy to give bounds on the same. The way we are going to give bounds, is to first estimate the partition function of the random cluster model with something simpler and then we derive our desired bounds from there. For this purpose, we first consider random cluster models with $q \in \mathbb{N}$ for the time being. Consider the random cluster model with $q \in \mathbb{N}$. Take $N = wI_q + J_q$. By coupling, that we proposed earlier, $Z_G(N) = Z_G(q, w)$.

Now, approximate N by a rank 1 matrix. The immediate choice of a rank 1 matrix, can be as follows, keeping the row sum and column sums intact: $M_1 = (1 + \frac{w}{q}) \mathbb{1}_q \mathbb{1}_q^T$, and we denote the “**rank 1 approximation**” of the random cluster model by $Z_G^{(1)}(q, w) := Z_G(M_1)$.

Now, we can show the following lemma stating that the rank 1 approximation leads to an under-approximation.

Lemma 1. *If $q \geq 1$, then,*

$$Z_G^{(1)}(q, w) \leq Z_G(q, w).$$

And if $q \in (0, 1)$, the inequality is reversed.

In rank 1 approximation, we fail to see the existence of an immediate upper bound. So, it is natural to move to rank 2 approximations. For that purpose, we are now approximating the matrix N with a rank 2 matrix, one such choice of the rank 2 approximator can be the

following, keeping the first row and column intact and approximating the remaining principal sub-matrix with a rank 1 matrix, i.e. approximate N by

$$M_2 = \begin{pmatrix} 1+w & 1 & \dots & 1 \\ 1 & 1+\frac{w}{q-1} & \dots & 1+\frac{w}{q-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1+\frac{w}{q-1} & \dots & 1+\frac{w}{q-1} \end{pmatrix}.$$

We define the rank 2 approximation of the partition function for integer values of the parameter q as follows:

$$Z_G^{(2)}(q, w) := Z_G(M_2).$$

For non-integer q . Now, the problem can be, in the above definition of rank 2 approximation, we have used a $q \times q$ matrix, and so, it is only valid for random cluster models with integer values of q only. To get over this, we note that, for

$$M'_2 = \begin{pmatrix} 1+w & 1 \\ 1 & 1+\frac{w}{q-1} \end{pmatrix} \quad \mu'_2 = \begin{pmatrix} 1 \\ q-1 \end{pmatrix},$$

the partition function of this 2-spin model is same as the partition function of the spin model indexed by M_2 , i.e. $Z_G(M_2) = Z_G(M'_2, \mu'_2)$. And as this definition does not use that q is an integer, so it can be extended for the cases of non-integer q 's too. And hence, we define the rank 2 approximation of the partition function of the (q, w) random cluster model as follows:

$$Z_G^{(2)}(q, w) = Z_G(M'_2, \mu'_2).$$

So, essentially, we have approximated the random cluster model by a suitable 2-spin model. Next we investigate the efficiency of the rank 2 approximator. It is shown in the following lemma:

Theorem 2 (Bounds for Rank 2 approximation). *For $q \geq 2$,*

$$Z_G^{(2)}(q, w) \leq Z_G(N, \underline{\mu}) \leq q^{n/g+L(g)} Z_G^{(2)}(q, w)$$

for all $g \in \mathbb{N}$. Recall, $L(g)$ is the number of cycles of size at most $(g-1)$, and n is the number of vertices of the graph.

Now we say that it is sufficient to check the sequence of the pressure function for these rank-2 approximations only, and if that converges, then the sequence of the pressure function of the original random cluster model also converges almost surely, for a sequence of d regular random graphs. Thus, we have the following theorem.

Theorem 3 ([1]). *Let $q \geq 2$ and $w > 0$. Let $(G_n)_n$ be a sequence of random d -regular graphs on n vertices. Consider the sequences of pressure functions for the rank 2 approximation and the random cluster model, respectively, i.e.,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln Z_{G_n}^{(2)}(q, w), \text{ and } \lim_{n \rightarrow \infty} \frac{1}{n} \ln Z_{G_n}(q, w).$$

If the first limit exists almost surely, then the second limit exists too almost surely, and they have the same value.

Proof of Theorem 3. By Theorem 2, we get that

$$\begin{aligned} Z_{G_n}^{(2)}(N, \underline{\mu}) &\leq Z_{G_n}(N, \underline{\mu}) \leq q^{n/g+L} Z_{G_n}^{(2)}(N, \underline{\mu}) \\ \implies \frac{1}{n} \ln Z_{G_n}^{(2)}(N, \underline{\mu}) &\leq \frac{1}{n} \ln Z_{G_n}(N, \underline{\mu}) \leq \left(\frac{1}{g} + \frac{L}{n} \right) \ln q + \frac{1}{n} \ln Z_{G_n}^{(2)}(N, \underline{\mu}), \end{aligned}$$

taking log each side and dividing it by the number of vertices n .

Now, as the sequence of d -regular graphs is a large girth sequence with probability 1, so, $\frac{L}{n}$ goes to 0, and this is true for every positive integer g , and as the limit exists, so we conclude that the sequence of the pressure function of random cluster on the random d -regular graphs also converge and converge to the same value. $\square$

Now we look at a clever technique to represent the parameter matrix.

Rank 2 matrix decomposition. Consider an r -spin model with parameters $(N, \underline{\mu})$, where N is an $r \times r$ matrix and $\underline{\mu} \in \mathbb{R}^r$.

Note that if N is a rank 2, positive definite matrix, there exist vectors a and b such that N can be decomposed as $N = aa^T + bb^T$.

Remark: For M'_2 , as that is also rank 2 and positive definite, such decomposition exists, and one might take the following decomposition:

$$\underline{a} = \begin{pmatrix} \sqrt{1 + \frac{w}{q}} \\ \sqrt{1 + \frac{w}{q}} \end{pmatrix} \quad \text{and} \quad \underline{b} = \begin{pmatrix} \sqrt{\frac{(q-1)w}{q}} \\ -\sqrt{\frac{w}{q(q-1)}} \end{pmatrix}.$$

These are the terms that appear in the trigonometric objective function, whereas, the following proposition explains the appearance of the sine and cosine functions in the same:

Proposition 1. *If (a, b) are a valid pair of decomposition for N , Then the following is also valid,*

$$a_j(t) = a_j \cos(t) + b_j \sin(t),$$

$$b_j(t) = -a_j \sin(t) + b_j \cos(t),$$

where $j \in \{1, 2, \dots, r\}$.

Now we take digressions to subgraph counting polynomials and then to belief propagation in the next section.

4. DIGRESSION: SUBGRAPH COUNTING POLYNOMIAL AND BELIEF PROPAGATION

4.1. Subgraph Counting Polynomial. For the later analysis we define the notion of subgraph counting polynomial (see [7]).

Definition 4. *Let G be a d -regular graph. Introduce the subgraph counting polynomial as follows :*

$$F_G(x_0, x_1, x_2, \dots, x_d) = \sum_{A \subseteq E} \prod_{v \in v} x_{\deg_A(v)},$$

where $\deg_A(v)$ denotes the the degree of the vertex v with respect to the edge set A . We can define more generally as :

$$F_G(x_0, x_1, x_2, \dots, x_d | z) = \sum_{A \subseteq E} \prod_{v \in v} x_{\deg_A(v)} z^{2|A|}.$$

It turns out that we can elegantly write the partition function of the $(N, \underline{\mu})$ spin model in the terms of a sub-graph counting polynomial for a d -regular graph. Next, we want to write the partition function in the terms of the sub-graph counting polynomial.

$$\begin{aligned}
Z_G(N, \underline{\mu}) &= \sum_{\sigma: V \rightarrow [r]} \prod_{v \in V} \underline{\mu}_{\sigma(v)} \prod_{\{u, v\} \in E} N_{\sigma(u), \sigma(v)} \\
&= \sum_{\sigma: V \rightarrow [r]} \prod_{v \in V} \underline{\mu}_{\sigma(v)} \prod_{\{u, v\} \in E} (aa^T + bb^T)_{\sigma(u), \sigma(v)} \\
&= \sum_{A \subseteq E} \sum_{\sigma: V \rightarrow [r]} \prod_{v \in V} \underline{\mu}_{\sigma(v)} \prod_{\{u, v\} \in E \setminus A} a_{\sigma(u)} a_{\sigma(v)} \prod_{\{u, v\} \in A} b_{\sigma(u)} b_{\sigma(v)} \\
&= \sum_{A \subseteq E} \prod_{v \in V} \left(\sum_{k=1}^r \underline{\mu}_k a_k^{d-d_A(v)} b_k^{d_A(v)} \right) \\
&= F_G(r_0, r_1, \dots, r_d)
\end{aligned}$$

where, $r_j = \sum_{k=1}^r \underline{\mu}_k \cdot a_k^{d-j} \cdot b_k^j$ for $j \in \{1, 2, \dots, d\}$.

We define $r_j(t)$ accordingly as $r_j(t) = \sum_{k=1}^r \underline{\mu}_k \cdot a_k(t)^{d-j} \cdot b_k(t)^j$. We also denote the quantity $r_0(t)$ by the notation $\Phi_{a, b, \underline{\mu}}(t)$.

$$(1) \quad \Phi_{a, b, \underline{\mu}}(t) = r_0(t).$$

In later section, we shall see the correspondence of this quantity with the limiting quantity $\Phi_{d, q, w}$.

4.2. Bethe Functional, Bethe Free Energy and Belief Propagation. In statistical physics, belief propagation (see [5] for details) is a recursive algorithm to calculate the partition function. It can be shown that the fixed-point solutions, in case of tree graphs, give the true value of partition function. It is hence believed that in case of locally tree-like graphs too, the fixed points correspond to a sensible (under-) estimate for the partition function, called Bethe free energy. The objective function comes from a functional by Bethe, calculated for d -regular graphs, which by some suitable transformation gives the trigonometric objective function and the set of fixed points gets reparameterized to the interval $[-\pi, \pi]$.

Bethe belief propagation recursion: In case of an $(N, \underline{\mu})$ spin model on random d -regular graphs, which have locally tree like structure, the Bethe belief propagation recursion is given by: $BP(h)_\sigma = \mu_\sigma \left(\sum_{\sigma'} N_{\sigma, \sigma'} h(\sigma') \right)^{d-1}$, where, h is a probability distribution in $\mathbb{R}^r$. Then we define the so-called Bethe functional for d -regular graphs.

Definition 5 (Bethe Functional).

$$\tilde{\Phi}_{d, N, \underline{\mu}}(h) = \log \left[\sum_{\sigma} \underline{\mu}_{\sigma} \left(\sum_{\sigma'} N_{\sigma, \sigma'} h_{\sigma'} \right)^d \right] - \frac{d}{2} \log \left[\sum_{\sigma, \sigma'} N_{\sigma, \sigma'} h_{\sigma} h_{\sigma'} \right].$$

To get our estimate for the partition function, we take this entropy term, and after exponentiating it, take its supremum over the set of fixed points of the recursion discussed above.

Definition 6 (Bethe Free Energy).

$$\Phi_d(N, \underline{\mu}) = \sup_{h \in \mathcal{H}} \exp(\tilde{\Phi}_{d,N,\underline{\mu}}(h))$$

where, $\mathcal{H}$ is the set of fixed points of the Bethe belief propagation recursion.

In the following section, we shall first calculate the limit and then, by using the subgraph counting polynomials and the belief propagation, conclude that the limiting quantity is the maximum value of a trigonometric objective function.

5. EVALUATION OF THE LIMIT AND TRIGONOMETRIC BETHE APPROXIMATION

In this section, we first discuss how we calculate the limit of the pressure function, and then we try to connect it with the trigonometric objective function appearing in the main result. We shall use these three theorems in this section to prove the main theorem of this section, Theorem 7.

Theorem 4 ([2]). *Let G be a random d -regular graph on n vertices. We consider the $(N, \underline{\mu})$ spin model on this sequence, where N is 2×2 symmetric positive definite matrix with positive entries and $\underline{\mu}$ is a vector in $\mathbb{R}^2$ with both entries positive. Then,*

$$\mathbb{E}[Z_G(N, \underline{\mu})] = n^{O(1)} \Phi_d(N, \underline{\mu})^n.$$

Theorem 5 ([6]). *For a 2×2 positive definite matrix N with positive entries and $\underline{\mu} \in \mathbb{R}_{\geq 0}^2$ and a d -regular graph G we have $Z_G(N, \underline{\mu}) \geq \Phi_d(N, \underline{\mu})^{|V(G)|}$.*

Convergence of the pressure function.

Theorem 6 ([1]). *For any essentially large girth sequence of d -regular graphs $(G_n)_n$, (recall Definition 3 and that $v(G_n) \rightarrow \infty$) the sequence of pressure functions, i.e.*

$$\frac{1}{v(G_n)} \ln Z_{G_n}(N, \underline{\mu})$$

converges.

Now, as we know by Theorem 6 that the limit exists, so to find the limit, it is sufficient to consider the sequence of d -regular random graphs $(G_n)_n$ with n vertices.

Theorem 7 ([1]). *Let N be a 2×2 positive definite matrix with positive entries and $\underline{\mu} \in \mathbb{R}_{>0}^2$. if $(G_n)_n$ is a sequence of random d -regular graphs with n vertices, then*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln Z_{G_n}(N, \underline{\mu}) = \ln \Phi_d(N, \underline{\mu})$$

with probability 1.

Proof of Theorem 7. By Theorem 4, $\mathbb{E}Z_{G_n}(N, \underline{\mu}) = n^C \Phi_d(N, \underline{\mu})^n$. Again, by Theorem 5, $Z_{G_n}(N, \underline{\mu}) \geq \Phi_d(N, \underline{\mu})^n$. Fix $\varepsilon > 0$.

$$\begin{aligned} & \left| \frac{1}{n} \ln Z_{G_n}(N, \underline{\mu}) - \Phi_d(N, \underline{\mu}) \right| > \varepsilon > 0 \\ \iff & \frac{1}{n} \ln Z_{G_n}(N, \underline{\mu}) - \left[C \frac{\ln n}{n} + \frac{1}{n} \ln \mathbb{E}Z_{G_n}(N, \underline{\mu}) \right] > \varepsilon \\ \iff & \frac{1}{n} \ln \frac{Z_{G_n}(N, \underline{\mu})}{\mathbb{E}Z_{G_n}(N, \underline{\mu})} > \varepsilon + C \frac{\ln n}{n} \end{aligned}$$

$$\begin{aligned} \iff \ln \frac{Z_{G_n}(N, \underline{\mu})}{\mathbb{E}Z_{G_n}(N, \underline{\mu})} &> n\varepsilon - C \ln n \\ \iff \frac{Z_{G_n}(N, \underline{\mu})}{\mathbb{E}Z_{G_n}(N, \underline{\mu})} &> \frac{e^{n\varepsilon}}{n^C} \quad (*) \end{aligned}$$

The probability of (*) by Markov is at most $\frac{n^C}{e^{n\varepsilon}}$ which is summable by ratio test as $\varepsilon > 0$. Hence, by Borel-Cantelli lemma, we conclude. $\square$

Now, we see in detail, how the quantity $\Phi_{d,q,w}$ in the main result comes. Recall 1 that, $\Phi_{a,b,\underline{\mu}}(t) = r_0(t)$.

Again, consider an $(N, \underline{\mu})$ 2-spin model where N is a 2×2 matrix with a decomposition (a, b) , hence, we get that

$$\Phi_{a,b,\underline{\mu}}(t) = \underline{\mu}_1 a_1(t)^d + \underline{\mu}_2 a_2(t)^d$$

. We take this trigonometric quantity and we maximize over the interval $[-\pi, \pi]$.

Definition 7.

$$\tilde{\Phi}_d(N, \underline{\mu}) = \max_{t \in [-\pi, \pi]} \Phi_{a,b,\underline{\mu}}(t).$$

We claim, in the next theorem that for this spin model, this quantity is actually same as the quantity $\Phi_d(N, \underline{\mu})$.

Theorem 8. *Let $N \in \mathbb{R}^{2 \times 2}$ be a positive definite matrix with positive entries and $\underline{\mu} \in \mathbb{R}_{>0}^2$. Then,*

$$\Phi_d(N, \underline{\mu}) = \tilde{\Phi}_d(N, \underline{\mu}).$$

For the proof, we first rewrite the Bethe fixed point equation, $\text{BP}(h) = h$ using the substitution $R = \frac{h_1}{h_2}$:

$$R = \frac{\underline{\mu}_1}{\underline{\mu}_2} \left(\frac{N_{11}R + N_{12}}{N_{12}R + N_{22}} \right)^{d-1}.$$

Call $\mathcal{R}_{N,\underline{\mu}}$ the set of non-negative solutions of this equation. By a simple calculation we have

$$\begin{aligned} \bar{\mathbb{F}}(R, N, \underline{\mu}) &:= \exp \left(\tilde{\Phi}_{N,\underline{\mu},d}(h) \right) \\ &= \left(\sum_{\sigma} \underline{\mu}_{\sigma} \left(\sum_{\sigma'} N_{\sigma,\sigma'} h_{\sigma'} \right)^d \right) \left(\sum_{\sigma,\sigma'} N_{\sigma,\sigma'} h_{\sigma} h_{\sigma'} \right)^{-\frac{d}{2}} \\ &= \underline{\mu}_1 \left[\frac{N_{11}R + N_{12}}{\sqrt{N_{11}R^2 + 2N_{12}R + N_{22}}} \right]^d + \underline{\mu}_2 \left[\frac{N_{12}R + N_{22}}{\sqrt{N_{11}R^2 + 2N_{12}R + N_{22}}} \right]^d. \end{aligned}$$

We know that

$$\Phi_d(N, \underline{\mu}) = \max_{R \in \mathcal{R}_{N,\underline{\mu}}} \bar{\mathbb{F}}(R, N, \underline{\mu}).$$

Now, let us also choose a representation $N = aa^T + bb^T$, and let

$$R(t) = -\frac{b_2(t)}{b_1(t)} \quad \text{and} \quad S(t) = \frac{a_1(t)}{a_2(t)}.$$

By simple calculation, we can show this following lemma:

Lemma 2. Let $a_1, a_2, b_1, b_2, \underline{\mu}_1, \underline{\mu}_2 \in \mathbb{R}$, then for every $t \in [0, 2\pi]$ such that $b_1(t), a_2(t) \neq 0$ we have

$$S(t) = \frac{N_{11}R(t) + N_{12}}{N_{12}R(t) + N_{22}}.$$

Furthermore,

$$|a_1(t)| = \frac{|N_{11}R(t) + N_{12}|}{\sqrt{N_{11}R(t)^2 + 2N_{12}R(t) + N_{22}}}$$

and

$$|a_2(t)| = \frac{|N_{12}R(t) + N_{22}|}{\sqrt{N_{11}R(t)^2 + 2N_{12}R(t) + N_{22}}},$$

and if t_0 maximizes $\Phi_{\underline{a}, \underline{b}, \underline{\mu}}(t)$, then

$$R(t_0) = \frac{\underline{\mu}_1}{\underline{\mu}_2} \left(\frac{N_{11}R(t_0) + N_{12}}{N_{12}R(t_0) + N_{22}} \right)^{d-1}.$$

Proof of Theorem 8. Take the decomposition so that $a_1, a_2, b_1, b_2 > 0$ (such a decomposition exists due to Proposition 1, considering the spectral decomposition and then using a suitable value of $t' \in [-\pi, \pi]$):

$$\Phi_{\underline{a}, \underline{b}, \underline{\mu}}(t) = \underline{\mu}_1 [a_1 \cos t + b_1 \sin t]^d + \underline{\mu}_2 [a_2 \cos t + b_2 \sin t]^d$$

By continuity of $\Phi_{\underline{a}, \underline{b}, \underline{\mu}}$, $\exists t_0 \in [0, \pi/2]$ such that $\tilde{\Phi}_d(N, \underline{\mu}) = \Phi_{\underline{a}, \underline{b}, \underline{\mu}}(t_0)$. Thus, by the previous lemma,

$$R(t_0) = \frac{\underline{\mu}_1}{\underline{\mu}_2} \left[\frac{N_{11}R(t_0) + N_{12}}{N_{12}R(t_0) + N_{22}} \right]^{d-1}$$

$\therefore R(t_0) \in \mathcal{R}_{N, \underline{\mu}}$. Now note that

$$\tilde{\Phi}_d(N, \underline{\mu}) = \Phi_{\underline{a}, \underline{b}, \underline{\mu}}(t_0) = \bar{\mathbb{F}}(R(t_0), N, \underline{\mu}) \leq \max_{R \in \mathcal{R}_{N, \underline{\mu}}} \bar{\mathbb{F}}(R, N, \underline{\mu}) = \Phi_d(N, \underline{\mu}).$$

Again, by continuity, $\exists t_1 \in [0, 2\pi]$ so that $R(t_1) = R_0$ where $\bar{\mathbb{F}}(R_0, N, \underline{\mu}) = \Phi_d(N, \underline{\mu})$.

Note that $S(t_1) = \frac{a_1(t_1)}{a_2(t_1)} = \frac{N_{11}R_0 + N_{12}}{N_{12}R_0 + N_{22}} > 0$. $\therefore a_1(t_1)$ and $a_2(t_1)$ have the same sign. Make both positive (Change t_1 to $t_1 + \pi$ if needed).

Thus, $a_1(t_1) = \frac{N_{11}R_0 + N_{12}}{\sqrt{N_{11}R_0^2 + 2N_{12}R_0 + N_{22}}}$ and $a_2(t_1) = \frac{N_{12}R_0 + N_{22}}{\sqrt{N_{11}R_0^2 + 2N_{12}R_0 + N_{22}}}$. Thus we get,

$$\Phi_d(N, \underline{\mu}) = \bar{\mathbb{F}}(R_0, N, \underline{\mu}) = \Phi_{\underline{a}, \underline{b}, \underline{\mu}}(t_1) \leq \tilde{\Phi}_d(N, \underline{\mu}).$$

Thus, we conclude the proof. $\square$

6. CONCLUSION

From the discussion in Theorem 8, we see that for any 2-spin model with a positive definite parameter matrix N , the limit of pressure function of the random cluster model on a sequence of random d -regular graphs, is same as the quantity $\ln \Phi_d(N, \underline{\mu})$. Now, we saw that this quantity can be identified as the maximum of the trigonometric function $\Phi_{\underline{a}, \underline{b}, \underline{\mu}}(t)$ in the

interval $[-\pi, \pi]$. Now, if we take $N = M'_2$, and take $\underline{a}, \underline{b}$ as $\left(\frac{\sqrt{1 + \frac{w}{q}}}{\sqrt{1 + \frac{w}{q}}} \right)$ and $\left(\frac{\sqrt{\frac{(q-1)w}{q}}}{-\sqrt{\frac{w}{q(q-1)}}} \right)$

respectively, and $\mu_1 = 1, \mu_2 = q - 1$, then the function

$$\begin{aligned} \Phi_{a,b,\underline{\mu}}(t) = & \left[\left(\sqrt{1 + \frac{w}{q}} \cos(t) + \sqrt{\frac{(q-1)w}{q}} \sin(t) \right)^d \right. \\ & \left. + (q-1) \left(\sqrt{1 + \frac{w}{q}} \cos(t) - \sqrt{\frac{w}{q(q-1)}} \sin(t) \right)^d \right]. \end{aligned}$$

Thus

$$\begin{aligned} \Phi_d(N, \underline{\mu}) = & \max_{t \in [-\pi, \pi]} \left[\left(\sqrt{1 + \frac{w}{q}} \cos(t) + \sqrt{\frac{(q-1)w}{q}} \sin(t) \right)^d \right. \\ & \left. + (q-1) \left(\sqrt{1 + \frac{w}{q}} \cos(t) - \sqrt{\frac{w}{q(q-1)}} \sin(t) \right)^d \right], \end{aligned}$$

and as this is a function of d, q, w , hence we denote this quantity by $\Phi_{d,q,w}$ in this context. Now, we can say by Theorem 7 and Theorem 8 that the limit of the pressure function of the 2-spin model with parameters (M'_2, μ'_2) , on a sequence of d -regular random graphs, converges to $\ln \Phi_{d,q,w}$. So, by Theorem 3, we conclude that the sequence of the pressure function of the (q, w) random cluster model on that sequence of random d -regular graphs also converges almost surely to the same quantity. Thus, we conclude our main result, Theorem 1.

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